MIL-A-23670A (WP)
1 NOVEMBER 1965
SUPERSEDING
MIL-A-23670 (WEP)
1 MAY 1963

MILITARY SPECIFICATION

ATTITUDE HEADING REFERENCE SYSTEM AN/ASN-50(*)

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - This specification covers design and all performance requirements for the Attitude Heading Reference System AN/ASN-50.

1.2 Classification - The attitude heading reference system shall consist of the items listed in 6.9.

2. APPLICABLE DOCUMENTS

2.1 General - The documents listed in 6.11 of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

*3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to Specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in Specification MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable. (See 6.4.)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

*3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in Specification MIL-E-5400.

*3.2.2 Electron Devices - Electron tubes, transistors and diodes shall be chosen and applied as outlined in Specification MIL-E-4682.

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*3.3 Design and Construction - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

*3.3.1 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 51.5 pounds. Component weights are:

(1)	Displacement Gyroscope	26.5 lbs.
(2)	Amplifier-Power Supply	8.3 lbs.
(3)	Compass Adapter	13.2 lbs.
(4)	Compass Controller	2.0 lbs.
(5)	Rate Switching Gyroscope	1.5 lbs.

3.3.2 Reliability

*3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of a least 20 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

*3.3.2.2 Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

*3.3.2.3 Reliability in Mean Time Between Failures (MTBF) - The equipment shall have 325 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.3.2.4 Time Totalizing Meter - The equipment shall contain time totalizing meters in accordance with Specification MIL-M-7793. Meters shall be included in the following units.

Unit	Type of Meter
Amplifier-Power Supply	MS17323
Displacement Gyroscope	MS17323
Compass Adapter	MS17322

3.3.3 Cabling and Connections -

*3.3.3.1 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400.

*3.3.3.2 Interconnection Cabling - The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the equipment.

*3.3.4 Control Panels - All rack or console mounted control panels shall conform to the applicable requirements of Specification MIL-C-6781. The configuration of all control panels must be approved by the procuring activity prior to preproduction testing.

*3.3.5 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-E-5400.

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*3.3.6 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

*3.3.7 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field shop maintenance.

*3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

*3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25°C ± 5°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0/200 ± 2.0 VAC and 27.5 ± 0.5 VDC

*3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class 2 equipment, except as modified below (see 4.2.2).

3.3.10.1 Vibration - The equipment, with the exception of the compass controller, shall operate satisfactorily when subjected to the tests of MIL-E-5272, Procedure XII, Curve A, (see 4.2.2.1). The compass controller shall operate satisfactorily when subjected to the tests of MIL-E-5272, Procedure V.

*3.3.10.2 Temperature-Altitude - Temperature shall be permitted to stabilize at 95°C.

3.3.10.3 Acceleration - The equipment shall operate satisfactorily when subjected to the tests of MIL-STD-810, Method 513.

3.3.10.4 Humidity - The equipment shall operate satisfactorily when subjected to the tests of MIL-E-5272, Procedure I.

3.3.10.5 Acoustical Noise - The gyroscope assembly shall be subjected to Procedure I of MIL-STD-810, grade B.

3.3.10.6 Rain - The equipment shall be subjected to the tests of MIL-E-5272, Procedure II.

*3.3.11 Warm-up Time - The time required for the equipment to warm-up prior to operations shall be kept to a minimum and shall not exceed 3.5 minutes under standard conditions and 5.0 minutes at extreme service conditions.

*3.3.12 Primary Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704 having limits as specified therein. The power required shall not exceed the specified amounts.

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(1) AC Power (Three-phase) 115/200V, Category "B" 200 VA.
(2) DC Power 28V, Category "B" 0.10 Amp.

*3.3.12.1 Degraded Performance - Degraded performance will be permitted for voltage transients not exceeding 0.5 second during normal electric system operation. Operation shall return to normal with no resulting damage to the equipment. (See 3.5.2.6.)

3.3.13 Cases - Cases of the components shall conform to the applicable figure of this specification, have non-ferrous, low density metal mounting flanges, and shall be finished with a lusterless black material Color No. 37038 of FED-STD-595.

3.3.14 Dielectric Breakdown - The materials used and the design shall be such that the resistance between isolated circuits and between each circuit and the case or connector shell is at least 50 megohms when a potential of 500 VDC is applied for a period of 10 seconds.

*3.4 Performance - The Attitude Heading Reference System shall provide gyroscopically stabilized roll, pitch, and heading reference data for aircraft. These shall be synchro type signals (3-wire).

*3.4.1 Attitude - The system shall provide aircraft roll and pitch outputs which are referenced to gravity. The outputs shall be gyro stabilized.

*3.4.2 Heading - The system shall provide aircraft heading outputs which are referenced to North heading. The heading output data shall be available from three modes of operation as selected by the pilot.

*3.4.2.1 Free Mode - In the free mode of operation, heading data shall be derived from a free running directional gyroscope. Provisions shall be made in the system to correct the data for gyro drift caused by earth's rate and by gyro drift bias.

*3.4.2.2 Slave Mode - In the slaved mode of operation, heading output data shall be slaved to earth's magnetic and shall be gyro stabilized. Earth's rate and gyro drift bias corrections shall be supplied in this mode.

*3.4.2.3 Compass Mode - In the compass mode of operation, heading data from the magnetic compass is repeated in the output without gyro stabilization. This is an emergency mode of operation intended to provide heading data only in case of gyro failure.

*3.4.3 Output Accuracies - The system, when connected and operated as a system at standard conditions, shall be capable of providing the statistical one sigma accuracies of output signals as indicated below:

Output	Accuracy
Roll and Pitch	0.10 degree
Magnetic Heading	0.10 degree
System Heading Drift, DG Mode	0.25 degree per hour
Vertical Gyro Drift	5.0 degree per hour

3.4.4 Performance at Environmental Extremes - Performance deviations at environmental extremes shall be permitted as specified in the Detail Requirements and when approved in the Test Procedures of paragraph 4.6.

3.4.5 System Interconnections - The component units when electrically interconnected as a system shall provide signals and control circuits as shown in Figure 1. The interconnecting wiring shall not be supplied as part of the system.

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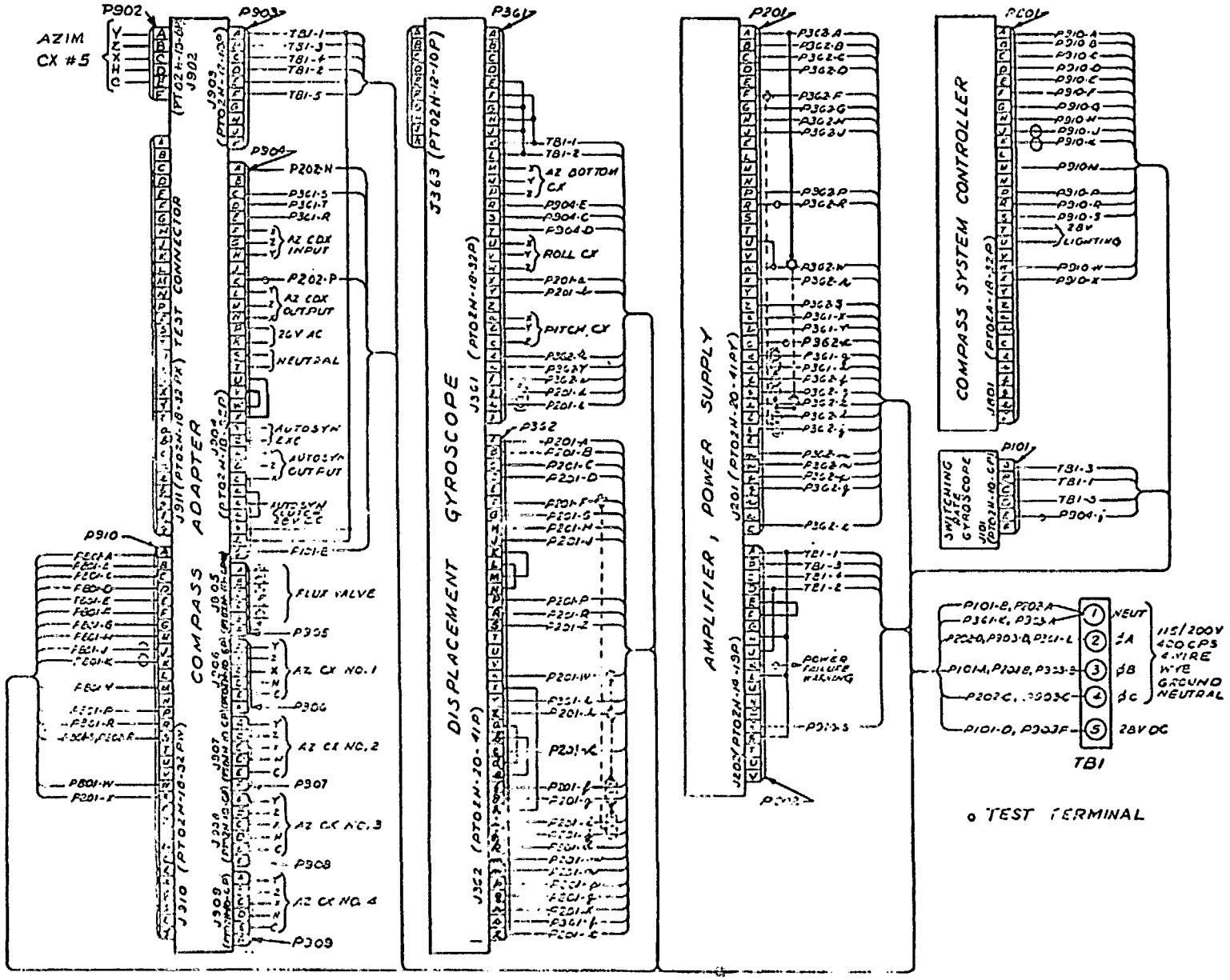


FIGURE 1. SYSTEM INTERCONNECT

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3.5 Detail Requirements -

3.5.1 Displacement Gyroscope - The gyroscope assembly shall be designed and constructed to meet the requirements of this specification. No gimbal binding under differential pressure conditions shall occur as a result of interference between case and gimbal should a leak develop in service. It shall be designed to operate with the Amplifier-Power Supply.

The displacement gyroscope assembly shall consist of a vertical gyro and a directional gyro mounted in multiple gimbals with a common outer gimbal as described in the following paragraphs. This gimbal arrangement is used in order to avoid the gimbal lock position. The outline dimensions of the assembly shall be in accordance with Figure 2.

3.5.1.1 Vertical Gyro - The vertical gyro shall have sufficient angular momentum and be of such design so as to provide the performance specified herein when operated throughout the temperature, vibration, and acceleration ranges specified.

3.5.1.1.1 Gimbal System - The vertical gyro shall be suspended in a gimbal system having two free inner gimbals and a servo-driven outer gimbal to eliminate gimbal lock of the gyro. The outer gimbal shall be used in common with the directional gyro.

3.5.1.1.1.1 First Vertical Gyro Gimbal - The first gimbal shall be the gyro rotor housing. It shall be stabilized by the inertia of the gyro.

3.5.1.1.1.2 Second Vertical Gyro Gimbal - The second gimbal shall be the next gimbal outward from the first gimbal. It shall be stabilized in pitch by the gyro.

3.5.1.1.1.3 Third Vertical Gyro Gimbal - The third gimbal shall be the outer gimbal. It shall be servo driven to maintain a fixed relation between the first and second gimbal except for certain conditions specified herein.

3.5.1.1.2 Vertical Gyro Axes of Freedom - The vertical gyro assy shall have three axes of freedom. Each axis shall be a signal axis.

3.5.1.1.2.1 First Vertical Gyro Axis of Freedom - The first axis of freedom shall be between the first and second gimbal. This axis shall be maintained in the horizontal plane and be aligned with the orthogonal horizontal projection of the fore and aft axis of the reference assembly under normal conditions. A minimum freedom of ± 80° shall be provided about this axis. An angular displacement about this axis shall provide an indication which changes from roll to yaw as the pitch angle changes from zero pitch to ± 90° pitch. Stops shall be incorporated on this axis which produce a pitch precession when the freedom is exceeded and no damage shall result under this condition.

3.5.1.1.2.2 Second Vertical Gyro Axis of Freedom - The second axis of freedom shall be between the second gimbal and the third (servo driven) gimbal. This axis shall be perpendicular to the first axis of freedom and shall be maintained in the horizontal plane under normal conditions. Unlimited freedom shall be provided about this axis. An angular displacement about this axis shall be a measurement of the pitch angle.

3.5.1.1.2.3 Third Vertical Gyro Axis of Freedom - The third axis of freedom shall be between the third gimbal (outer roll) and the instrument frame or case. This axis shall be perpendicular to the second axis of freedom and shall coincide with the fore and aft axis of the instrument. Unlimited freedom shall be provided about this axis. An angular displacement about this axis shall be a measurement of the roll angle.

3.5.1.1.3 Gimbal Actuation - The inner two gimbals are maintained in a fixed relation to the apparent gravity vertical by the gyro erection system. The outer roll gimbal is actuated by a servo mechanism and is driven so as to maintain the spin axis of the vertical gyro and the pitch axis in a 90°

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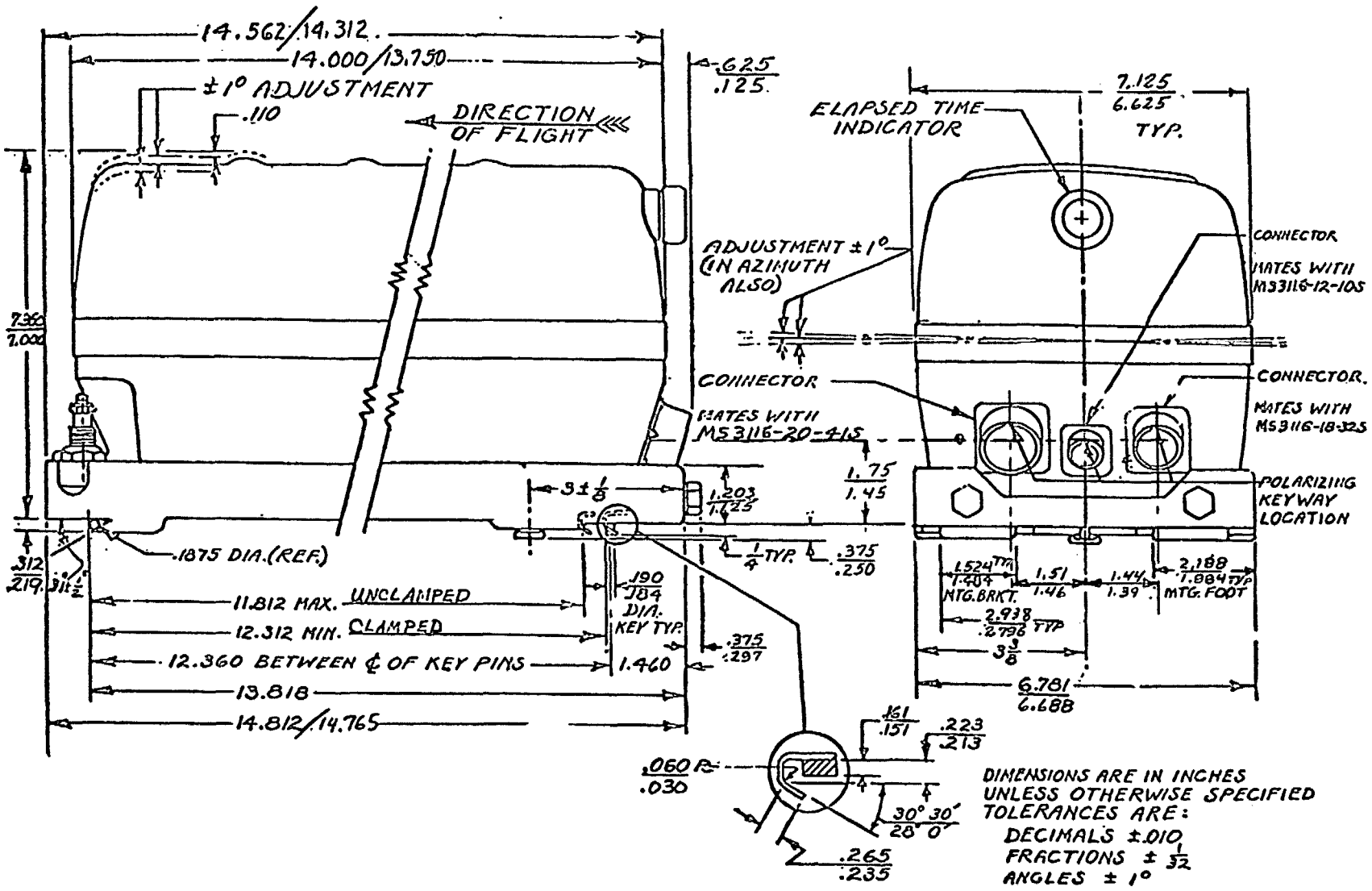


FIGURE 2. DISPLACEMENT GYROSCOPE ASSEMBLY

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relation to each other except for certain transient conditions. The response of the servo of the outer roll gimbal shall be such that this 90° relation will be maintained during roll rates from zero through 360° per second with min. lag, and no overshooting, oscillation, hunting, or jumping of such a magnitude and frequency which will affect the performance of a high performance autopilot. The servo loop shall be tight enough so that during loop maneuvers when the outer roll gimbal is not controlled from external commands the outer roll gimbal rotates 180° in order to keep proper sensing of the servo drive when the outer roll gimbal axis passes through the vertical.

3.5.1.1.3.1 Special Outer Roll Gimbal Operation - The design of the circuitry of the outer roll gimbal servo shall be such that by external connection, the outer roll gimbal will perform as required under 3.4.1.1.3 or when connected to an external control, eg. bombing computer, the outer roll gimbal will be controlled in response to commands from the external control in order to provide continuous outputs during certain aircraft maneuvers.

In a typical maneuver, the bombing computer will resolve inner roll and azimuth signals such that when the outer roll gimbal is near vertical the gyro azimuth signal will control the outer roll gimbal thus giving an effective roll signal about the aircraft axis. During this maneuver, the outer gimbal will be stabilized so that it does not rotate 180° about its axis as the reference is rotated in pitch through or near vertical. The outer roll gimbal shall be controlled to keep the pitch axis in a plane which is continuously parallel to the outer roll gimbal axis and perpendicular to a vertical plane which is parallel to the heading held at the time the operation was initiated. The control shall be limited such that the outer roll gimbal axis must remain within ± 20° of the vertical plane above.

3.5.1.1.4 Erection - The erection mechanism shall incorporate separate systems for pitch and roll. Erection shall be accomplished by electrolytic switches and torque motors on the gimbals. When the vertical gyro is displaced in one axis, it shall not deviate in the other axis while it is erecting by more than 1.0°. The erection system shall erect the gyro from any attitude to the vertical as determined by the vertical sensing electrolytic switches. The pitch and roll erection torquer leads shall be brought out of the case so that erection cut-out may be provided and external voltages may be applied to precess the gyro in pitch and roll. Proper erection voltages shall be obtained from the Amplifier-Power Supply to provide the following erection characteristics.

3.5.1.1.4.1 Initial Erection - With the proper input signals the vertical gyro shall have fast initial erection so that it assumes a position within 1° of vertical in pitch and roll within 1-1/2 minutes, or less, after power is applied.

3.5.1.1.4.2 Final Erection and Verticality - The vertical gyro shall align its spin axis with the direction of gravity within ± .25° in 2 minutes after the initial erection period.

3.5.1.1.4.3 Normal Erection Rate - The rate of normal erection about the roll axis shall be nominally 1.3° per minute and shall fall between the limits plotted on Figure 3 throughout all environments and roll attitudes of the control case. The rate of normal erection about the pitch axis shall be nominally 1.3° per minute and shall fall between the limits plotted on Figure 3 throughout all environments. Pitch erection sensing shall be maintained when displacing the gyro in roll ± 45° or less.

3.5.1.1.4.4 Fast Erection - With the proper input signal, the gyro shall be capable of fast erection to the apparent vertical in both axes of the gyro simultaneously. The fast erection rate shall not be less than 15° per minute in either axis at room temperature or 12° per minute at temperature extremes.

3.5.1.1.4.5 Roll and Pitch Pendulum Erection Cutout - The pitch and roll erection torquer leads shall be brought out of the case to provide a means to cut out pitch erection, roll erection, or both simultaneously. Pendulum erection cutout shall not affect the torquing of the gyro resulting from input of plane and earth rate correction.

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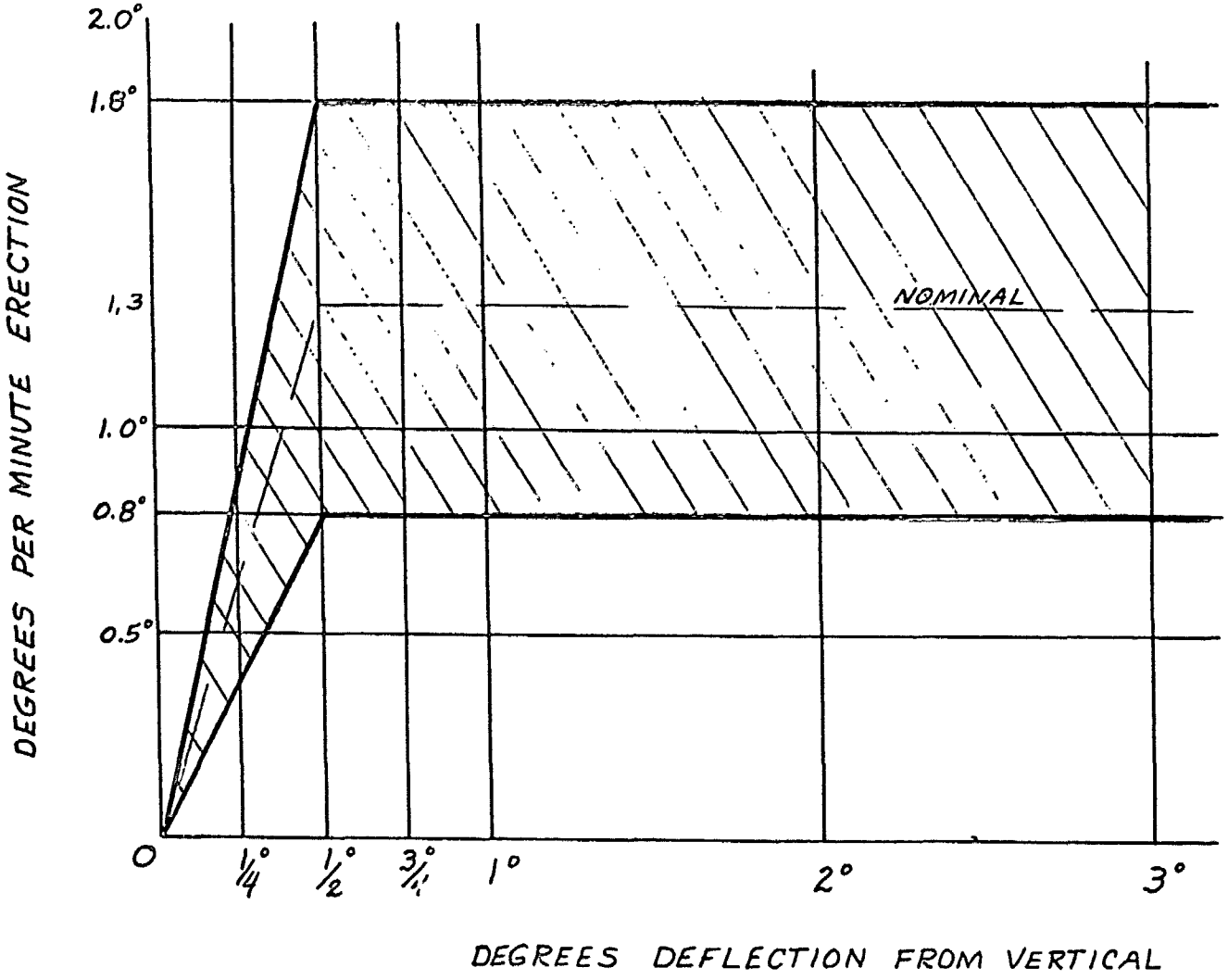


FIGURE 3. ERECTION RATE AS A FUNCTION OF DEFLECTION

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3.5.1.1.4.6 Vertical Gyro Correction - Provision shall be made to apply torques in pitch and roll in either direction. The design shall be such that the torquers will work directly from either an amplifier or transformer. The torquers shall be internally excited such that when approximately 15 VAC, 400 cps, phase C +90° is applied to the control phase legs, the torquer shall cause the gyro to precess at a nominal rate of 1° ± .2°/minute. The control phase impedance shall be 350 ± 70 ohms + J260 + 60 ohms. The power consumption in each torquer fixed phase shall not exceed 1 watt.

*3.5.1.1.4.6.1 Earth Profile Correction - Provision shall be made within the gyroscope for earth profile corrections corresponding to Mach 0.6 or Mach 1.2. This shall be accomplished such that application of 26 VAC, phase C + 90°, shall torque the vertical gyro in pitch at a rate of .115 ± .05° per min. or .23 ± .05° per min. respectively. Interconnections shall be such that torque rates of 3.4.1.1.4.6 are not affected if no connection is made.

*3.5.1.1.5 Vertical Gyro Synchro Outputs - Two transmitting synchros shall be mounted on each vertical gyro axis of freedom except inner roll which shall have one. These synchros shall provide displacement signals of gimbal movement about their respective axes at a one to one ratio. One each of the synchros on the pitch and outer roll axes shall have their stator and rotor leads brought out to the signal electrical connector, Table I, separate from each other and other circuitry. These synchros shall each be capable of operating up to 8 high impedance control transformers each having an impedance of 220 ± 11 + J470 ± 24 ohms. The synchros shall have a nominal no load line-to-line output of 11.8 ± 0.25V and phase shift of 4.0 ± 1.5°. When loaded with 8 CT's, the line-to-line output voltage shall not be less than 10.6V and the phase shift shall not be more than 10°.

The remaining synchros on the vertical gyro axes of freedom shall be internally excited with the voltage across terminals A and D of the control electrical connector, Table II. The stator leads of the synchros shall be brought out through the output connector terminals in accordance with Table I; however, they may be connected to other control circuitry provided up to 6 high impedance control transformers having characteristics as defined above can be operated externally from each. The internally excited pitch and outer roll synchro transmitters shall have a lead from the stator center tap brought out to connector terminals (Table III) for use in failure detection circuits.

3.5.1.1.5.1 Synchro Zeroing and Sensing - The transmitting synchros on the vertical gyro axes of freedom shall be zeroed and their lead terminals of Table I connected to satisfy the procedures of 3.5.1.1.5.1.1 through 3.5.1.1.5.1.5 for the identification of the terminals. The angular spread between the nulls of the two synchros on the outer roll axis (and on the pitch axis) shall not exceed 0.4°. The null voltages shall not exceed 30 millivolts.

3.5.1.1.5.1.1 Externally Excited Pitch Synchro - Terminals c and K Table I shall be connected to the power lead terminal A, Table II and terminal L of Table I shall be connected to the power lead terminal D, Table II. The voltage across terminals a and b, and the voltage across terminals L and a or b of Table I shall be measured with the gimbals at zero displacement. The voltage across terminals a and b of Table I shall be at the minimum and the voltage across L and a or b of Table I shall be less than the voltage applied to terminals A and D of Table II. The voltage across terminals a and c and across b and c of Table I shall be measured as the reference assembly is rotated in climb about the pitch gimbal. The voltage across terminals a and c, Table I shall increase before decreasing and the voltage across the terminals b and c, Table I shall decrease before increasing.

3.5.1.1.5.1.2 Externally Excited Outer Roll Synchro - Terminals W and H of Table I shall be connected to the power lead terminal A of Table II, and terminal J of Table I shall be connected to the power lead terminal D of Table II. The voltage across terminals U and V of Table I and the voltage across terminals J and U or V of Table I shall be measured with the gimbal at zero displacement. The voltage across terminals U and V of Table I shall be at the minimum and the voltage across J and U or V of Table I shall be less than the voltage applied to terminal A and D of Table II. The voltage across terminals U and W and across V and W of Table I shall be measured as the reference assembly is rotated about the outer roll gimbal in right roll. The voltage across terminals U and W shall increase before decreasing and the voltage across the terminals V and W shall decrease before increasing.

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TABLE I

Displacement Gyroscope Assembly Output Connector (J361)

Bendix P/N PT02H-18-32P, or equal, to mate with MS3116P-18-32S or equal

Terminal	Function
A	Spare
B	Spare
C	Spare
D	Roll Switch - 90° - 270°
E H Rotor	Azimuth CX, ØC, TOP
F C Rotor	Azimuth CX, BOTTOM
G H Rotor	115 VAC excit. - F to G in phase with "X" to "Z" at zero displacement
H C Rotor	Roll CX - DG
J H Rotor	115 VAC excit. - H to J in phase with "Z" to "X" at zero roll
K C Rotor	Pitch CX - EXT.
L H Rotor	115 VAC excit. - K to L in phase with "Z" to "X" at zero pitch
M X Stator N Y Stator P Z Stator	Azimuth CX - BOTTOM
R X Stator S Y Stator T Z Stator	Azimuth CX - TOP
U X Stator V Y Stator W Z Stator	Roll CX - DG
X X Stator Y Y Stator Z Z Stator	Roll CX - VG (Internally excited rotor - neutral to ØC in phase with "Z" to "X" at zero roll)
a X Stator b Y Stator c Z Stator	Pitch CX - EXT
d X Stator e Y Stator f Z Stator	Pitch CX - INT (Internally excited rotor - neutral to ØC in phase with "Z" to "X" at zero pitch)
g X Stator h Y Stator j Z Stator	Inner roll CX (Internally excited rotor - neutral to ØC in phase with "Z" to "X" at zero displacement

3.5.1.1.5.1.3 Inner Roll Synchro - Terminal j of Table I shall be connected to terminal A of Table II. The voltage across g and h of Table I and the voltage across terminal D of Table II and terminal g or h of Table I shall be measured with the gimbals at zero displacement. The voltage across terminals g and h shall be at the minimum and the voltage across terminal D of Table II and terminals g or h of Table I shall be less than the voltage applied to terminals A and D of Table II. The voltage across terminals g and j and across h and j of Table I shall be measured as the first gimbal (inner roll) is held and the reference assembly with the outer roll gimbal is rotated in right roll. The voltage across terminals g and j of Table I shall increase before decreasing and the voltage across terminals h and j shall decrease before increasing.

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TABLE II

Displacement Gyroscope Assembly Input Connector (J362)

Bendix P/N PT02H-20-41P, or equal, to mate with MS3116P-20-41S or equal

Terminal	Function
A (Neutral)	Input Power - 3Ø 400 cps, 4 wire
B (ØA)	200V line to line/115V line to
C (ØB)	neutral
D (ØC)	ØA, ØB, and ØC phase rotation
E	Spare
F	Leveling signal
G	Leveling excitation
H	Leveling torquer control Ø
J	Leveling torquer fixed Ø
K	Pitch torquer A
L	Pitch torquer B
M	Pitch switch A
N	Pitch switch B
P	Pitch switch common
R	Pitch capacitor
S	Power failure
T	Pitch segment switch
U	Slaving torquer fixed Ø
V	Slaving torquer control Ø
W	Pitch servo CT (H)
X	Reserved
Y	Reserved
Z	Pitch generator output (c)
a	Roll switch B
b	Roll switch A
c	Roll torquer excitation
d	Roll torquer B
e	Roll torquer A
f	Roll generator output (hot)
g	Roll generator output (cold)
h	Pitch generator output & pitch servo CT (comm)
i	Leveling signal (A)
j	Leveling signal (B)
k	Reserved
m	Pitch M. G. control Ø
n	Pitch M. G. fixed Ø
p	Pitch generator excitation
q	Roll M. G. control Ø
r	Roll M. G. fixed Ø
s	Reserved
t	Roll generator excitation

3.5.1.1.5.1.4 Internally Excited Pitch Synchro - Terminal f of Table I shall be connected to terminal A of Table II. The voltage across d and e of Table I and the voltage across terminals D of Table II and terminal d or e of Table I shall be measured with the gimbals at zero displacement. The voltage across terminals d and e shall be at the minimum and the voltage across terminals D of

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TABLE III

Displacement Gyroscope Assembly Calibration Connector (J363)

To mate with MS3116P-12-10S or equal

Terminal	Function
A	Pitch torquer CT
B	Pitch torquer
C	Pitch torquer
D	Roll torquer CT
E	Roll torquer
F	Roll torquer
G	Roll Synchro Stator CT
H	Pitch Synchro Stator CT
J	Spare
K	Earth Profile Resistor

Table II and terminals d or e of Table I shall be less than the voltage applied to terminals A and D of Table II. The voltage across terminals d and f and across e and f of Table I shall be measured as the second gimbal (pitch) is held and the reference assembly with the outer roll gimbal is rotated in climb. The voltage across terminals d and f of Table I shall increase before decreasing and the voltage across terminals e and f shall decrease before increasing.

3.5.1.1.5.1.5 Internally Excited Outer Roll Synchro - Terminal Z of Table I shall be connected to terminal A of Table II. The voltage across X and Y of Table I and the voltage across terminal D of Table II and terminal X or Y of Table I shall be measured with the gimbals at zero displacement. The voltage across terminals X and Y shall be at the minimum and the voltage across terminal D of Table II and terminal X or Y of Table I shall be less than the voltage applied to terminal A and D of Table II. The voltage across terminals X and Z and across Y and Z of Table I shall be measured as the third gimbal (outer roll) is held and the reference assembly is rotated in right roll. The voltage across terminals X and Z of Table I shall increase before decreasing and the voltage across terminal Y and Z shall decrease before increasing.

*3.5.1.1.6 Random Drift - With normal distribution the random drift of the vertical gyro shall not exceed 5° per hour - one sigma in the roll and pitch axes, respectively, with the gyro spin axis at vertical.

3.5.1.1.6.1 Pendulosity Drift - To assure proper balance, the vertical gyro assembly shall be so designed that the maximum free drift, noted after compensation for the earth's rotation, shall not be greater than 15° per hour when operated over the entire temperature range specified herein, with the gyro spin axis tilted 20° of vertical.

3.5.1.1.7 Power-Off Freedom - When power is removed momentarily, the vertical gyro shall still have freedom about its first and second gimbal axes and shall not be displaced from the vertical during roll angles of ± 75° or pitch angles through ± 60° during this power-off condition.

3.5.1.1.8 Roll Segment Switch - Roll segment switch shall be incorporated on the third gimbal axis. This switch shall close terminal D of Table I to neutral from +90° through 180° to 270° of roll where zero roll is the initial zero position determined by 3.5.1.1.5.1.5. The circuit shall be open from the 270° roll position through zero roll to the +90° roll position.

3.5.1.2 Directional Gyro - The directional gyro shall be suspended in a gimbal system of which the outer gimbal is the third (servoed) gimbal of the vertical gyro. The directional gyro shall have three other gimbals suspended from this outer gimbal.

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3.5.1.2.1.1 Inner Directional Gyro Gimbal - The first (inner) directional gyro gimbal (gyro wheel housing) shall be stabilized by the inertia of the gyro. Its pivot axis shall be the first directional gyro axis of freedom and shall be perpendicular to the spin axis of the gyro and shall normally lie in the horizontal plane. The slaving torquer shall act about this axis. This gimbal shall have a minimum of ±80° freedom about its axis.

3.5.1.2.1.2 Directional Gyro Azimuth Gimbal - The second directional gyro gimbal shall be stabilized by the inertia of the gyro. Its pivot axis aligned with the vertical. The azimuth output synchros and the leveling torquer shall act about this axis. This gimbal shall be capable of continuous rotation about its axis.

3.5.1.2.1.3 Directional Gyro Pitch Gimbal - The third directional gyro (pitch) gimbal shall be servo driven so that it follows the vertical gyro second (pitch) gimbal at a maximum rate of not less than 200° per second. Its pivot axis shall be the third directional gyro axis of freedom and shall be parallel to the vertical gyro second axis (pitch) of freedom. This gimbal shall be capable of continuous rotation about its axis.

3.5.1.2.2 Directional Gyro Leveling - The inner gimbal shall be slaved by means of the leveling torquer controlled by a suitable nonpendulous sensor so that the spin axis of the directional gyro is perpendicular to the azimuth axis. This leveling rate shall have an initial rate which will level the gyro within 1-1/2 minutes after power is applied. It shall have a normal leveling rate of 2° to 6° per minute.

3.5.1.2.3 Directional Gyro Slaving - The directional gyro shall contain an azimuth torquer to provide an azimuth slaving capability with a rate of 1° to 2° per minute with application of the proper signal.

*3.5.1.2.4 Azimuth Transmitting Synchros - Two synchros shall be attached to the azimuth axis of the gyro to transmit signals proportional at a one to one ratio of the position of the gyro. The synchros shall not exceed a maximum transmission error of 8 minutes of arc. The synchros shall be of the three phase wye connected stator and salient pole single phase rotor types. The angular spread between the nulls of the two azimuth output synchros shall not exceed 0.4°, and the null voltage of each synchro shall not exceed 30 millivolts. The azimuth transmitter synchros shall be capable of operating up to eight high impedance control transformers each having an impedance of 220 ± 11 ohms + J470 ± 24 ohms. The synchros shall have a nominal no load line-to-line output of 11.8 ± 0.25V and phase shift of 4.0 ± 1.5°. When loaded with 8 CT's, the line-to-line output voltage shall not be less than 10.6V and the phase shift shall not be more than 10°.

3.5.1.2.4.1 Internally Excited Azimuth Transmitter - The stator leads of the internally excited synchro shall be brought out to the terminals of the signal connector in accordance with Table I and shall be separate from other circuitry. The synchro shall be internally connected to neutral and shall be excited by the voltage applied to terminal E per Table I.

3.5.1.2.4.2 Externally Excited Azimuth Transmitter - The transmitter shall operate from a rotor excitation of 115V 400 cps and the maximum stator output voltage, line to line, shall be 11.8V. The transmitter shall have an electrical accuracy of ± 8 minutes or less. The rotor and stator leads shall be brought to separate pins of the electrical connector per Table I, and shall be isolated from other circuitry.

3.5.1.2.4.3 Synchro Zeroing and Sensing - Terminal T of Table I shall be connected to terminal A of Table II. The voltage across R and S of Table I, and the voltage across D of Table II and R and S of Table I shall be measured with the gimbals at zero displacement, the spin axis of the directional gyro aligned with the fore aft axis of the displacement gyroscope assembly, and with the directional gyro spinning CW (clockwise) when viewed from the aft end. The voltage across terminals R and S of Table I shall be at the minimum and the voltage across terminal D of Table I and terminal R or S of Table I shall be greater than the voltage applied to terminals A and D of Table II. The voltage across terminals R and T and across S and T of Table I shall be measured as the directional gyro

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second (azimuth) gimbal is held and the reference assembly is rotated about its axis in right turn. The voltage across terminals R and T of Table I shall increase before decreasing and the voltage across terminals S and T of Table I shall decrease before increasing. When conducting this procedure pin E of Table I shall be connected to pin D of Table II.

3.5.1.2.4.3.1 Externally Excited Azimuth Transmitter - Terminals P and F of Table I shall be connected to the power lead terminal A of Table II, and terminal G of Table I shall be connected to the power lead terminal D of Table II. The voltage across terminals M and N of Table I and the voltage across terminals G and M or N of Table I shall be measured with the gimbals at zero displacement, the spin axis of the directional gyro aligned with the fore-aft axis of the reference assembly and with the directional gyro spinning CW when viewed from the aft end. The voltage across terminals M and N of Table I shall be at the minimum and the voltage across terminal D of Table II and terminals M or N of Table I shall be greater than the voltage applied to terminals A and D of Table II. The voltage across terminals M and P and across N and P of Table I shall be measured as the directional gyro second (azimuth) gimbal is held and the reference assembly is rotated about its azimuth axis in right turn. The voltage across terminals M and P of Table I shall increase before decreasing and the voltage across terminal N and P of Table I shall decrease before increasing.

3.5.1.2.5 Balance - The directional gyro shall be balanced so that the apparent drift due to earth's rotation is equal to the product of the sine of the latitude of operation and 15° per hour in a CCW (counterclockwise) direction in the Northern Hemisphere and in a CW direction in the Southern Hemisphere.

*3.5.1.2.6 Random Drift - With normal distribution the random drift of the directional gyro shall not exceed ±0.20° per hour.

3.5.1.2.7 Compensatable Mean Drift - The mean drift of the directional gyro shall not exceed ±1.5° per hour from the computed apparent drift for the latitude of operation over the temperature range from -54°C to +71°C.

3.5.1.2.8 Pitch Segment Switch - A pitch segment switch shall be incorporated on the second gimbal axis (pitch servoed gimbal axis). This switch shall cause continuity to exist between pins T and A of J362 at all pitch gimbal angular displacements exceeding 90 ±2° in either climb or dive from its normal level (zero) position and shall provide open circuit at all displacements less than 90 ± 2°.

3.5.1.3 Mounting - The mounting of the displacement gyroscope assembly shall be in accordance with Figure 2 and such that the displacement reference will not deflect by more than ±1/4° in pitch and roll or 0.02° in azimuth as a result of accelerations or any vibration between 0 and 18 cps, with a minimum amplitude of 0.020 inch. The torsional resonant period shall be greater than 18 cps.

3.5.1.3.1 Release - Means shall be provided for quick release of the reference assembly including mount from the aircraft.

3.5.1.3.2 Adjustment - Means shall be provided for precision adjustment of the reference assembly in relation to its mount.

3.5.1.4 Starting Time - The gyro rotors shall attain full speed from rest within 5 minutes or less after application of power.

3.5.1.5 Hermetic Sealing - The design of case for the displacement gyroscope shall provide hermetic sealing such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the unit may be subjected and such that it will pass the case leak test.

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3.5.1.5.1 Filling Medium - The filling medium shall be of at least 98 percent purity, free of dust particles and containing not more than 0.006 milligrams of water vapor per liter (due point -65°C) at the filling pressure. The filling medium shall be 95 percent helium and 5 percent carbon dioxide. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere or 1/2 atmosphere as required. Other gases may be used with prior approval of the procuring activity.

3.5.1.5.2 Case Leak - The sealed case shall have a leak rate that shall not cause more than 10% loss or contamination of the filling medium after 1,000 hours at approximately 1 atmosphere differential pressure.

3.5.1.6 Slip Ring Continuity - Slip rings between gimbals shall be designed to provide positive and continuous contact during aircraft maneuvers. Under the conditions listed below, no sharp peaks or discontinuities shall occur in the synchro output signals when loaded with a servo test indicator having line-to-line impedance of 90 ± 15% + J570 ± 15% ohms, or equivalent.

a. Rotating the reference case approximately 2 rpm in azimuth.

b. Tilting the reference case approximately ± 80° in pitch.

c. Rotating the reference case approximately 2 rpm in roll, except that roll outputs shall also meet requirements at 1/2 rpm.

3.5.1.7 Power Failure Signal - Loss of any phase of AC power shall cause a minimum of 15 volts to appear between the center-tap of the vertical gyro motor and power neutral. The center-tap of the vertical gyro motor shall be connected to a terminal of J362 (see Table II).

3.5.1.8 Performance at Environmental Extremes - When tested at environmental extremes, the displacement gyroscope shall meet the erection, verticality, slaving, leveling, VG pendulosity, and DG mean drift requirements except as noted below.

3.5.1.8.1 Initial Erection - After one minute of operation the roll erection error shall not exceed 1.5°.

3.5.1.8.2 Final Erection and Verticality - The final erection and verticality requirements shall be met within 5 minutes after the start cycle.

3.5.1.8.3 Output Signals - The output signals for roll, pitch, and azimuth shall not vary more than 1° during vibration error tests.

3.5.2 Amplifier-Power Supply - The Amplifier-Power Supply shall be designed and constructed to meet the requirements of this specification. The assembly shall be designed to operate with the gyroscope assembly. The outline dimensions shall be in accordance with Figure 4.

The amplifier-power supply unit shall fulfill the necessary servo and power requirements for proper normal gyroscope operation. It shall contain the amplifiers for the servoed gimbals and related equipment and the starting and erection controls. Its connectors and terminals functions shall be in accordance with Tables IV and V.

3.5.2.1 Amplifier Channels - The amplifier channels shall be interchangeable, modular, plug-in-type where practicable and where reliability is not reduced by such design requirements. The amplifiers shall not incorporate electronic tubes. Parallel circuits shall be used where practicable so that partial failure does not result in total loss of operation.

3.5.2.2 Starting Sequence Control - The starting sequence control shall provide the signals to properly erect and align the gyros. These signals are provided when the servoed gimbals

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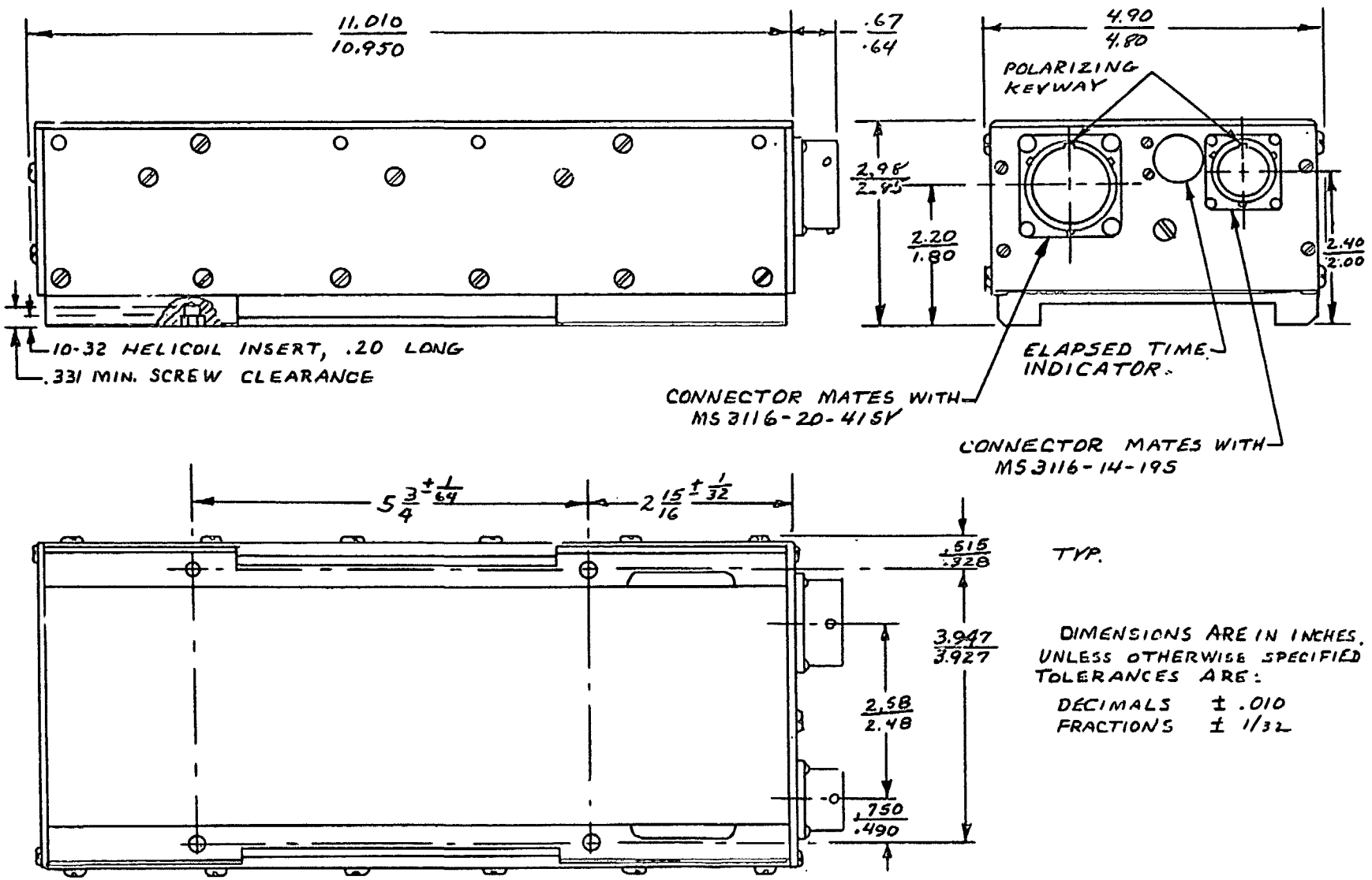


FIGURE 4. AMPLIFIER-POWER SUPPLY

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TABLE IV

Amplifier-Power Supply Input Connector (J202)

To mate with MS3116-14-19S or equal

Terminal	Function
A (Neutral)	115/200 VAC, 400 cps
B (ØA)	3 phase, 4 wire WYE
C (ØB)	Phase rotation A-B-C
D (ØC)	
E	Roll alarm - to F
F	Roll alarm - to E
G	Spare
H	Reserved
J	Reserved
K	Failure Warning
L	Failure Warning
M	Pitch erect cutout
N	Roll erect cutout
P	Cutout excit. - 115V, 400 cps, ØB
R	Start 60 sec. Auto sync - to S
S	Start 60 sec. Auto sync - to R
T	Reserved
U	Fast erection - to V
V	Fast erection - to U

are in any initial position and whenever the system power has been disconnected long enough to cause recycling, even though the gyros may still be coasting. Proper alignment shall also occur when the servoed shafts and gimbals of the systems are rotated to any new position after power is disconnected.

3.5.2.2.1 0 to 12 seconds - During the first 12 seconds after power turn on, the amplifier-power supply shall provide reduced voltage for starting the gyro motors, cut-off leveling and roll erection voltages, provide voltage for fast erection in pitch, open the power failure circuit, cage the outer roll gimbal, and provide synchronizing signal to the compas adapter.

3.5.2.2.2 12 to 60 seconds - During this period the amplifier-power supply shall provide full voltage to gyro motors, provide leveling voltage, provide voltage for fast erection in roll and pitch, provide synchronizing signal to the compass adapter, open the power failure circuit, and cage the outer gimbal.

3.5.2.2.3 After 60 seconds - After 60 seconds the amplifier-power supply shall provide all necessary voltages for continuous operation of the gyroscope.

3.5.2.3 Erection Control - The erection control shall provide the functions of fast erection control, roll erection cutout, and pitch erection cutout. External leads shall be provided to permit Roll and Pitch erection cutout and fast erection voltage when commanded by external switches.

3.5.2.3.1 Fast Erection - Fast erection voltage of 50 volts, phase C, shall be provided during the first 60 seconds after power turn-on, or when commanded by external circuits.

3.5.2.3.2 Normal Erection - Unless fast erection is commanded the erection voltage shall be 15 volts.

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TABLE V

Amplifier-Power Supply Output Connector (J201)

To mate with MS3116-20-41SY or equal

Terminal	Function	
A (Neutral) B (ØA) C (ØB) D (ØC)	Power , 400 cps 3 phase 4 wire WYE Phase rotation A-B-C	Output to Gyroscope
E	Test - Roll amp out	
F	Leveling input	
G	Leveling excitation	
H	Leveling control phase	
J	Leveling fixed phase	
K	Disable alarm relay	
L	Test - Roll driver out (amp in)	
M	Test - Leveling modulator out (amp in)	
N	Test - Alarm monitor	
P	Pitch erection voltage	
R	Pitch erection voltage	
S	Test - Unfiltered DC	
T	Test - Filtered DC	
U	Pitch alarm	
V	Test - Pitch amp out	
W	Pitch servo CT (H)	
X	Roll M-G fixed phase (w/o third gimbal controller)	
Y	Test - Leveling amp out	
Z	Power failure	
a	Roll servo input	
b	Start cycle - 60 sec.	
c	Roll erection voltage	
d	Roll servo input (w/o third gimbal controller)	
e	Roll servo input	
f	Roll servo input	
g	Start cycle - 60 sec.	
h	Pitch generator output (signal ground)	
i	Leveling input	
j	Leveling input, common	
k	Spare	
m	Pitch M-G control phase	
n	Pitch M-G fixed phase	
p	Pitch generator excitation	
q	Roll M-G control phase	
r	Roll M-G fixed phase (w/third gimbal controller)	
s	Test - Roll input transformer	
t	Roll generator excitation	

3.5.2.3.3 Erection Cutout - Separate control circuits shall be provided for cutting off roll and pitch erection voltages when commanded by external signals. Erection cutout from external signals shall be inoperative when fast erection is operated.

3.5.2.4 Servo Amplifiers - The amplifier-power supply shall provide signals for ROLL, PITCH, and LEVELING, to the gyro.

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3.5.2.4.1 Roll Servo Amplifier - The roll servo amplifier shall provide output to input voltages as follows:

Input Volts RMS	Output Volts RMS	Phase Relation
Input shorted	1 Max.	---
0.10 ± 0.002	21 ± 7	180° ± 30°
0.35 ± 0.007	28 Min.	---

3.5.2.4.2 Pitch Servo Amplifier - The pitch servo amplifier shall provide output to input voltages as follows:

Input Volts RMS	Output Volts RMS	Phase Relation
Input shorted	2.5 Max.	---
0.060 ± 0.002	10.0 ± 4	0° ± 20°
0.350 ± 0.010	14.0 Min.	0° ± 20°

3.5.2.4.3 Leveling Amplifier - The leveling amplifier shall provide output to input voltages as listed below: The output phase shall be either 0° ± 30° or 180° ± 30° with respect to ØC of input power depending on polarity of the input and as required to properly torque the gyro.

Input Volts DC	Output Volts RMS
0.050 ± 0.002	10.5 ± 4
0.250 ± 0.005	20.0 Min.

3.5.2.5 Failure Warning - Failure warning signals shall be provided for Power, Roll Servo, and Pitch Servo Alarms. These signals shall provide same output as Power Failure Warning.

3.5.2.5.1 Power Failure Warning - A 3-phase warning circuit shall be incorporated in the amplifier. This circuit shall operate a set of normally open contacts which shall be separate from other circuitry. When proper power is applied to the control, the contacts will close after the initial erection cycle is completed. If at any time any phase of the 3-phase power to the control is broken, the contacts will open.

3.5.2.5.2 Servo Failure - The amplifier-power supply shall have provisions for detecting an uncorrected error signal in the outer roll or pitch servo amplifiers, and for operating the power failure warning when an error is present. The failure alarm circuits shall be operative when externally connected to the amplifier inputs. An alarm disable relay shall be provided to disable the servo failure alarms when commanded by an external 28 VDC signal.

3.5.2.6 Power Interruption - Design of the amplifier-power supply shall be such that momentary interruption of power for periods less than 40 seconds will not cause the starting circuits to recycle.

3.5.2.7 Mounting - The mounting of the unit shall be such that the case can either be connected directly to the airframe or permit a mounting plate and/or vibration isolator to be inserted between the case and the mounting. Mounting dimensions are shown on Figure 4. When subjected to the high temperature test conditions of MIL-E-5400, the unit shall be mounted on a heat sink capable of dissipating 35 watts with a temperature differential of 3.5° C between the unit and the chamber air.

3.5.2.8 Performance at Environmental Extremes - When tested at environmental extremes, the amplifier-power supply shall repeat within 20%, the results of its individual tests made at standard conditions prior to the start of environmental tests, except that the servo amplifier maximum outputs shall be not less than 80% of specified outputs at standard conditions.

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3.5.3 Adapter Compensator, Compass - The compensator compass adapter shall furnish azimuth heading outputs. It shall receive its basic inputs from the ML-1 Compass Transmitter, per MIL-T-19576, the directional gyro, and the compass controller. Dimensions shall be in accordance with Figure 5. It shall provide the following functions: Azimuth (Directional Gyro) follow-up, magnetic slaving, heading synchronization, and DG corrections. Its connectors and terminal functions shall be in accordance with Table VI through XI.

*3.5.3.1 Azimuth Servo (DG Follow-Up) - The azimuth follow-up shall repeat the directional gyro heading output with high response and accuracy. This shall be direct output from the compass adapter. The azimuth servo system shall provide the following performance:

Static Accuracy	0° 15' RMS	or 0° 30' Max
Null Repeatability	0° 20'	
Follow-up Rate, Minimum	180°/sec.	
Follow-up Lag, Maximum	0.2°/RPM to 5 RPM	
Natural Frequency, Undamped	5 CPS min.	

*3.5.3.1.1 Dynamic Accuracy - The heading output shall not change more than 0.25° when operating with the mag. comp. oscillating up to ± 5° at 6 cpm.

TABLE VI

Compass Adapter to Controller Connector (J910)

To mate with MS3116P-18-32SW or equal

Terminal	Function
A	Latitude Compensation Supply
B	Latitude Compensation Supply
C	Compass Mode Relay
D	Slave Mode Relay
E	Latitude Compensation Signal
F	Latitude Compensation Signal
G	Neutral
H	Slew Motor Signal
J	Slew Pot Supply
K	Slew Pot Supply
L	Spare
M	Slew Relay
N	Spare
P	Sync. Indicator Signal
R	Sync. Indicator Signal
S	Sync. Relay
T	Spare
U	Spare
V	Spare
W	Man. Slew Autopilot Decouple
X	Man. Slew Autopilot Decouple
Y	Spare
Z	Spare
a thru j	Spares

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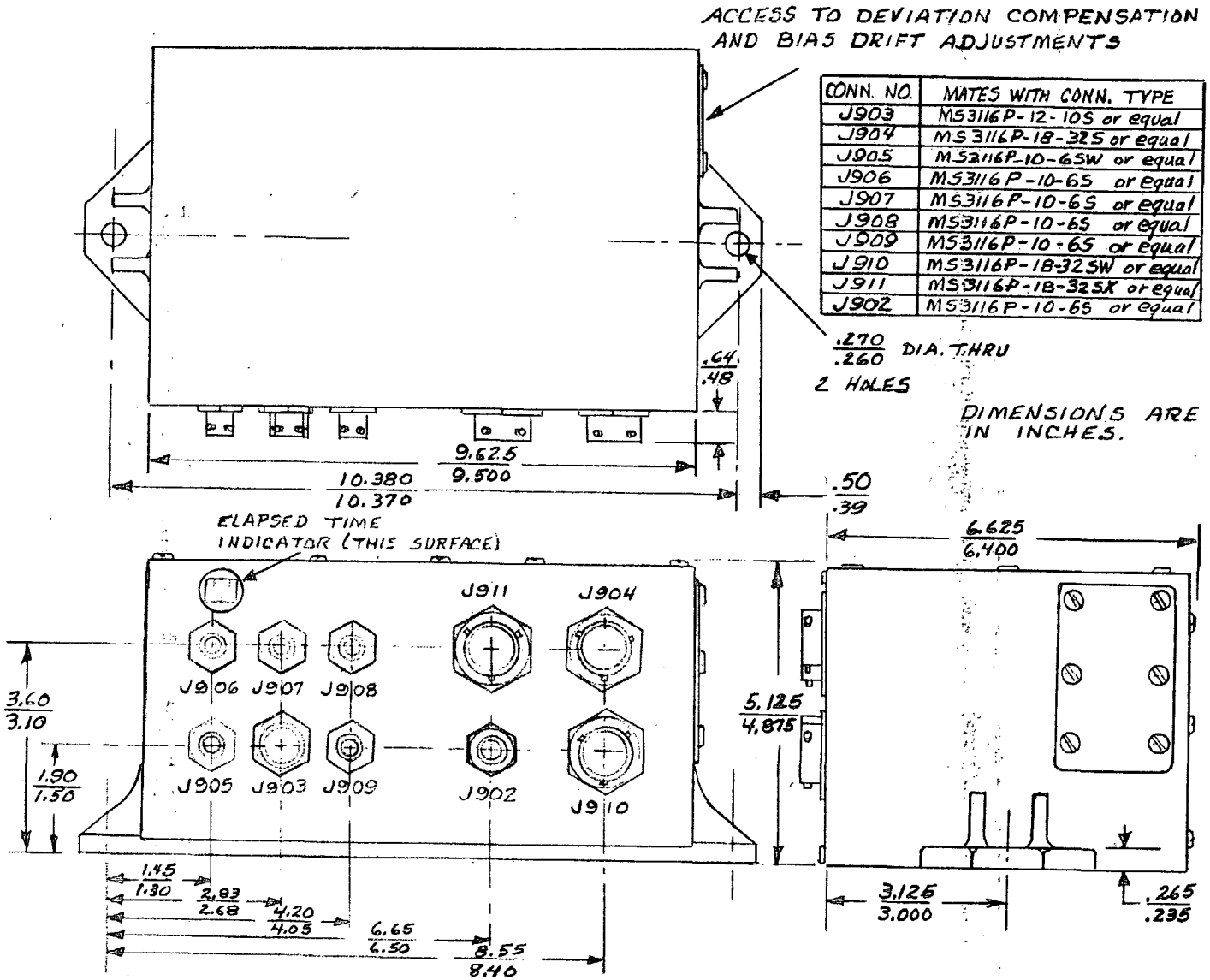


FIGURE 5. ADAPTER-COMPENSATOR, COMPASS

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TABLE VII

Compass Adapter Flux Valve Connector (J905)

To mate with MS3116P-10-6SW or equal

Terminal	Function
A	Compass CT input
B	Compass CT input
C	Compass CT input
D	Compass Transmitter excit.
E	Compass Transmitter excit.
F	Shield

TABLE VIII

Compass Adapter, CX Connectors (J902 and J906 thru J909)

To mate with MS3116P-10-6S or equal (5 req'd)

Terminal	Function
A	CX Output Y
B	CX Output Z
C	CX Output X
D	CX Excitation H
E	CX Excitation C
F	Spare

*3.5.3.1.2 Reversing Azimuth Servo Null - Reversal of the gyro CT output leads to the azimuth amplifier shall cause a 180° rotation of the 4 -, 8 -, and 20-power CX's but shall not reverse the differential synchro output.

3.5.3.2 Correction and Slaving Servo - The correction and slaving servo shall perform the functions of slaving the outputs of the compass adapter to the magnetic compass transmitter and provide correction signals for the heading outputs corresponding to earth's rate (latitude) and directional gyro drift bias.

3.5.3.2.1 Magnetic Slaving - Magnetic slaving shall slave the output of the compass adapter to the magnetic compass transmitter. The slaving signals shall provide the following performance:

Static Accuracy	0° 30' max.
Follow-up Rate	.7 to 1.5° per min.

3.5.3.2.2 Deviation Compensation - A 24 point electrical deviation compensator shall be provided to minimize errors of the remote compass transmitter and its associated control system. It shall only be effective when the magnetic compass signal is being utilized. The 24 point deviation compensator shall provide compensation with minimum range of ± 3.5 degrees at each adjustment point.

The deviation compensator shall be a plug in module and be easily accessible. The compass adapter compensator shall contain provisions which will permit elimination of the deviation compensation signal from the slaving signal as an aid in magnetic compass compensation.

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TABLE IX

Compass Adapter Signal Connector (J904)

To mate with MS3116P-18-32S or equal

Terminal	Function	
A	Erection cutout	
B	Spare	
C	Gyro CT input	Y
D	Gyro CT input	Z
E	Gyro CT input	X
F	Azimuth CDX input	X
G	Azimuth CDX input	Z
H	Azimuth CDX input	Y
J	Erection cutout	
K	Erection cutout	
L	Azimuth CDX output	Y
M	Azimuth CDX output	Z
N	Azimuth CDX output	X
P	CX excitation source	H
R	CX excitation source	H
S	CX neutral source	C
T	CX neutral source	C
U	Gyro CT output	
V	Gyro CT output	
W	Azimuth amplifier input	
X	Azimuth amplifier input	
Y	Clutched CX excitation	
Z	Clutched CX excitation	
a	Clutched CX output	Y
b	Clutched CX output	Z
c	Clutched CX output	X
d	Clutch	
e	Clutch	
f	Autopilot cutout	
g	Turn cutout relay	
h	Autopilot cutout	
j	Turn cutout relay	

3.5.3.2.3 Automatic Gain Control - The Compass Adapter shall contain automatic gain control to maintain slaving accuracy and slaving rate requirements when supplied with signal from an ML-1 magnetic compass operating in a magnetic field with horizontal component varying between 0.08 and 0.40 oersteds.

*3.5.3.2.4 Heading Corrections - Signals shall be available in slave and free modes to compensate for the DG mean drift up to 5° per hour and for the apparent DG drift caused by earth's rotation up to 15° per hour. The correction servo shall integrate these signals and provide a controlled rate of correction under standard conditions as follows:

Range	-	0.5° to 20° per hour minimum
Random Error	-	± 0.15° per hour rms.
Maximum Error	-	± 0.3° per hour

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TABLE X

Compass Adapter Power Input Connector (J903)

To mate with MS3116P-12-10S or equal

Terminal	Function
A (Neutral)	Input power - 3Ø, 400 cps
B (ØA)	4 wire, 200V line-to-line/115V
C (ØB)	Line to neutral, ØA, ØB, ØC rotation
D (ØC)	
E	DC ground
F	+28 VDC
G	
H	Spares
J	
K	

TABLE XI

Compass Adapter Test Connector (J911)

To mate with MS3116P-18-32SX or equal

Terminal	Function
A	Tachometer Output
B	Tachometer Output
C	Spare
D	AGC Input
E	Reserved
F	Remodulated 400 cps
G, b	Slave and Correction Amplifier Output
H	Ground
J	Pre-Amplifier-Limiter Output, and Slaving and Correction Amplifier Input
K, e	Azimuth Amplifier Output
L	Azimuth Amplifier Input
M	800 cps Slave Preamp. Output
N	Common
P	AGC DC Control
R	Deviation Compensation
S	Earth's Rate Correction
T	Gyro Drift Correction
U	Slaving and Correction Motor VG
V	Common
W thru j	Spares

3.5.3.3 Synchronization - Means shall be provided to automatically synchronize the directional gyro input signal to the magnetic compass transmitter at a fast rate during an initial power application, during switching to slave mode, or by manually depressing the push-to sync button on the controller. Synchronization shall be at a minimum rate of 20°/sec. and shall be accurate to ± 2°.

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3.5.3.3.1 Indicator Signal - A synchronization signal shall be provided for driving the compass controller synchronizing indicator meter. This signal shall provide a current of 10 ±3 microamperes per degree (1100 ohm load) and shall be limited to a maximum of 250 microamperes. Polarity of the signal shall indicate the direction of the error. The signal at null shall not exceed ±5 microamperes.

3.5.3.4 Set Heading Control - The Compass Adapter in combination with the controller shall provide for setting of heading while in "FREE" mode of operation. The output heading data from the Compensated Compass Adapter shall increase (+) when the knob of the controller is rotated clockwise. Speed of this rotation shall be adjustable from 1 to 9 RPM as determined by how far the Set Heading Knob is rotated.

*3.5.3.5 Data Transmitting Synchros - The outputs of the compass adapter shall be obtained from the following:

a. One 20-power transmitter
b. Two 4-power transmitters
c. One differential transmitter
d. One clutched synchro
e. Two 8-power transmitters

Items a, b, d and e shall be driven by the azimuth servo, and item c shall be driven by the correction and slaving servo.

*3.5.3.6 Synchro Requirements - Each 4-power transmitter shall be capable of driving two receiver synchros each having input impedance of 45 ± 3 ohms + J255 ± 13 ohms, or four control transformers each having input impedance of 220 ± 11 ohms + J470 ± 24 ohms. The synchros shall have a nominal no load line-to-line output of 11.8 ± 0.35V and phase shift of not more than 11.5°. When loaded with 4 CT's, the line-to-line output voltage shall not be less than 10.6V and the phase shift shall not be more than 17°. Each 4-power synchro shall have a transmission accuracy of ± 7' or less. The 20-power transmitter shall be capable of suppling 10 receivers or 20 control transformers under the foregoing condition. Each 8-power transmitter shall be capable of driving 4 receivers or 8 control transformers as above.

3.5.3.6.1 Clutched Synchro - The clutched synchro shall be excited thru a fixed resistor in series with the rotor. Synchro and resistor shall be such that, when the output from 2 stator leads (blue to blk or yel to blk) are loaded with 5000 ± 10% ohms and the synchro rotor (with resistor) are excited with 26 ± .5V, 400 ± 8 cps; the maximum output shall be 10.2 ± 1.2 volts at a phase angle of 30° ± 2° lead.

3.5.3.7 Synchro Zeroing - The output synchros shall be set at electrical zero for North (0°) heading output. The differential synchro shall be zeroed such that when the rotor is excited by a synchro transmitter operating at its electrical zero, the stator output will be at its electrical zero + 180°.

3.5.3.7.1 With the CX (J908) at 0° and the transmitter at 0° ±.5° and slaving cutout, the output of the CX's (J909, and J902) shall be 0° ± 12', and the output of the CX's (J906 and J907) shall be 0° ± 18'. The output of the CDX shall be 180° ± 30'.

NOTE: The heading synchro is adjusted for zero when the voltage from X to Y is at a minimum and the dial reading is 0° (NORTH), and with rotor lead C connected to stator lead Z the voltage from H to X shall be greater than the voltage from H to C. The voltage between X and Z shall increase before decreasing for a rotation of the dial in increasing heading direction. For decreasing heading rotation of the dial, the voltage between X and Z shall decrease before increasing.

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3.5.3.7.2 Clutched Synchro Zero - The output of the clutched synchro shall be 0 ± 1° with the clutch de-energized. With the clutch energized by 28 VDC the output shall increase for increasing heading and decrease for decreasing heading change up to ± 25°.

3.5.3.8 Compass Mode Operation - In the Compass Mode Operation, the output data transmitters follow the Magnetic Compass transmitter input. (No directional gyro stabilization.) The compass mode shall be initiated by an external signal from the mode selector switch. Static accuracy in the compass mode shall be ± 1° 30'.

3.5.3.9 Performance at Environmental Extremes - When tested at environmental extremes, the compass adapter shall meet the positioning accuracy, positioning rate, and synchronizing requirements except as indicated below. Adjustment of deviation potentiometers shall not be required during environmental tests.

3.5.3.9.1 Azimuth Servo - The free mode positional tolerances shall be increased from ± 30' max. to ± 45' max. and dead zone from 20' to 30'.

3.5.3.9.1.1 Compass Mode - The positional tolerances shall be increased from 1°30' to 2°.

3.5.3.9.2 Magnetic Slaving - The rate shall be increased from 3.5 and 7.5 degrees in 5 minutes to 2.5 and 8.5 degrees in 5 minutes. The positional accuracy shall be increased from 30' to 1.0°.

3.5.3.9.3 Synchronization - The positional tolerances shall increase from ± 2° to ± 2.5°.

3.5.3.9.4 Synchronization Indicator - The current through the 1100 ohm resistor shall increase from 5 microamperes or less to 8 microamperes or less. The current through the 1100 ohm resistor for a 5 degree displacement of the magnetic compass transmitter shall increase from 10 ± 3 microamperes to 10 ± 4 microamperes.

3.5.3.9.5 Set Heading Control - The minimum speed of 9 RPM shall be decreased to 8 RPM.

3.5.3.9.6 Heading Corrections - The maximum error shall be ± 0.5° per hour and the random error shall be ± 0.20° per hour.

3.5.3.10 Excitation Voltage Output - The compass adapter shall provide the following excitation voltages:

(1) 26 V., 400 cps, for excitation of synchro transmitters. The output shall be capable of simultaneously exciting the transmitters of para. 3.5.3.5 at room temperatures.

(2) 23.5 V., 400 cps, for excitation of the type ML-1 magnetic compass transmitter.

(3) 26 V., 400 cps, for excitation of the latitude compensation resolver in the compass controller.

(4) 3-0-3 V. (6.0 V CT), 400 cps, for excitation of the heading set potentiometer in the compass controller.

3.5.3.11 Cutout Relay - A 28 VDC relay shall be provided which, when energized, will interrupt the slaving operation of 3.5.3.2.1. The relay shall also interrupt a gyro erection circuit routed through the compass adapter.

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3.5.3.12 Autopilot Decoupling - A circuit routed through the compass adapter shall be interrupted whenever either manual or automatic synchronization occurs. The interruption shall occur prior to the start of synchronization.

3.5.3.13 Mechanical Requirement - The servomechanism assembly shall be modular in construction and be of the plug-in type.

*3.5.4 Controller, Compass System - The compass controller shall be designed and constructed to meet the requirements of this specification. Outline dimensions shall be in accordance with MS17392. It shall operate in conjunction with the compass adapter defined by paragraph 3.5.3.

The compass system controller shall provide the mode selection and controls for the operation of a directional gyro-magnetic compass heading system.

3.5.4.1 Controls - The compass system controller shall provide the following controls as shown on MS17392:

3.5.4.1.1 Mode Selector Switch - A switch shall be provided to select one of the following modes: Free, Slaved and Compass. The mode selector switch shall provide a circuit to operate compass and slave mode relays in the Compass Adapter. A second set of contacts shall be used to operate the synchronizing indicator in compass and slave modes.

3.5.4.1.2 Set Heading Control - The set heading control shall be labeled "PUSH TO TURN", shall provide switching to de-couple the autopilot and to control the direction and rate of slewing the system to an azimuth heading when operating in the "FREE" mode. The control shall consist of a Push-to-Turn switch and potentiometer. The switch shall, when operated, provide one circuit to operate the slew circuits of the compass adapter, and a second circuit to open autopilot cut-out.

3.5.4.1.2.1 Potentiometer - The set heading potentiometer shall have nominal resistance of 1000 ohms. The resistance between the fixed center tap and each end shall be equal within 80 ohms.

3.5.4.1.3 Synchronizing Switch - The synchronizing switch shall be labeled "PUSH TO SYNC" and shall provide for synchronization of the azimuth heading output to the magnetic compass when operating in the "SLAVED" mode.

3.5.4.1.4 Latitude Control - The latitude control shall provide means to compensate for apparent drift of the gyroscope resulting from rotation of the earth. With 26 ± 2.6 VAC applied and a load of 17.5K ohms ± 1%, the ratios between the in-phase component of the latitude control voltage and the excitation voltage shall be as specified below for each latitude setting:

LATITUDE SETTING	VOLTAGE RATIO
90°	.777 ± 1%
75°	.750 ± 1%
60°	.672 ± 1%
45°	.549 ± 1%
30°	.388 ± 1%
15°	.201 ± 1%
0°	.005 (max.)

For the "N" position of the hemisphere switch, the latitude control voltage shall lead the excitation voltage by 8° ± 3° and lag by 172° ± 3° for the "S" position of the hemisphere switch.

3.5.4.1.5 Hemisphere Switch - The hemisphere switch shall enable selection of the hemisphere in which latitude correction is to be applied.

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3.5.4.2 Control Knobs - The control knobs and marking thereon shall be in accordance with MS17392.

3.5.4.3 Synchronizing Indicator - The compass system controller shall include a synchronizing indicator to provide a visual indication of synchronization between the azimuth heading output and the magnetic compass. With the mode selector in the "SLAVED" or "COMPASS" position, 45 microamperes DC shall cause a minimum pointer deflection of 2/3 of full scale from its center position. Deflection shall be to the left when a left turn is required to synchronize.

3.5.4.3.1 Dial Marking - The synchronizing indicator dial shall be marked as shown on MS17392. The "L" and "R" and zero index markings shall be lusterless black, Color No. 37038 of FED-STD-595 and the scale shall be lusterless white, Color No. 37875 of FED-STD-595.

3.5.4.4 Panel Illumination and Marking - The compass system controller shall contain a lighting plate conforming to Specification MIL-P-7788. The plate shall be marked in accordance with Specification MIL-M-18012.

3.5.4.5 Mounting - Mounting provisions shall be in accordance with MIL-C-6781, and MS17392.

3.5.4.6 Connection - The electrical connector and terminal functions shall be in accordance with Table XII.

3.5.4.7 Performance at Temperature Extremes - The compass system controller shall perform as specified herein at temperature extremes except as follows:

3.5.4.7.1 Voltage Ratio - The voltage ratio tolerances of ± 1% in paragraph 3.5.4.1.4 shall be ± 2%.

3.5.4.7.2 Phase Angle - The phase angle tolerances of ± 3° in paragraph 3.5.4.1.4 shall be ± 5°.

3.5.4.7.3 Illumination - Illumination shall be checked to note that lighting is present, but no quantitative measurements shall be required.

3.5.5 Gyroscope, Rate Switching - The rate switching gyroscope shall be in accordance with MIL-G-23723. Dimensions shall conform to Figure 6.

The gyroscope shall be designed to provide switching during turn. The gyroscope shall consist of a rate gyro and associated turn-switching circuits.

*3.6 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (see paragraph 6.7).

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TABLE XII

Compass Controller Assembly Connector (J801)

To mate with MS3116E-18-32S or equal

Terminal	Function
A	Latitude compensation excitation
B	Latitude compensation excitation
C	"COMPASS" mode control signal
D	"SLAVE" mode control signal
E	Latitude compensation signal
F	Latitude compensation signal
G	Neutral
H	Slew control signal
J	Slew control excitation
K	Slew control excitation
L	Spare
M	Slew mode control signal
N	Spare
P	Synchronizing indicator
R	Synchronizing indicator
S	Synchronizing switch
T	Lighting - 28V
U	Lighting - Ground
V	Spare
W	Autopilot decouple
X	Autopilot decouple
Y	Spare
Z	Spare
a	Spare
b	"FREE" mode control
c	"FREE" mode control
d	Spare
e	Cosine latitude signal
f	Cosine latitude signal
g	Spare
h	Spare
j	Spare

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Tests

*4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

*4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422, except as modified below and in 3.3.10, and interference tests in accordance with Specification MIL-I-6181.

4.2.2.1 Vibration - The vibration tests for the displacement gyroscope, amplifier-power supply, and the compass adapter shall be divided into three equal periods to be run at -54°C, +71°C, and room temperature with level of curve A to 5 g at temperature extremes. Vibration at resonance shall be 10 minutes in each axis at each temperature.

*4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the

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same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

*4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

*4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

*4.3.2 Accessory Material - In addition to the complete equipment submitted for Initial Production Tests the contractor shall also submit such accessory material and data necessary to test the equipment.

*4.3.3 Initial Production Sample Approval - Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

*4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

*4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

*4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test

*4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

*4.4.2 Sampling Tests - Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	0*
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

*NOTE: If by contract action the Initial Production Test (4.3) is deleted, then a Sample Test shall be conducted on one equipment from the first 10 produced.

*4.4.2.1 Scope of Tests' - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the pre-production, initial production, individual and special tests.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by Specification MIL-R-23094. Equipments selected for reliability assurance tests shall first have passed the individual tests.

*4.4.3.1 Reliability Procedure - For the Reliability Qualification Plan, as outlined in Specification MIL-R-23094, Procedure I shall be used. For the Reliability Sampling Plan Procedure I shall be used.

*4.4.3.2 Test Level - Except for the initial temperature cycle of +71°C (+160°F) followed by -54°C (-65°F), the high temperature shall be +55°C (130°F) and the low temperature shall be -40°C (-40°F). The initial temperature cycle shall be conducted at the start of each system and on each unit thereafter upon repair or replacement. The length of each cycle shall not exceed 24 hours nor shall either of the high or low temperature conditions be less than 4 hours. The equipment shall be vibrated at 0.5 g at a non-resonant frequency between 20 and 60 cps for 10 minutes at each extreme temperature. No unit shall be subjected to temperature or vibration conditions exceeding the unit specification requirements.

*4.4.3.2.1 Initial Cycle - Upon receipt of the equipment, following operational tests (or debugging), the system shall be placed in a chamber at +71°C (+160°F) and operated for 4 hours using nominal voltage. The system shall be tested by Scorsby motion during this test (at least ±5°). The system shall then be tested at -54°C (-65°F) after a 4-hour soak at -54°C (-65°F).

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*4.4.3.3 Accept-Reject Criteria - Figure 13 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the Reliability Qualification Test Plan. Figure 14 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the Reliability Sampling Test Plan.

*4.4.3.4 Duty Cycle - The system shall be operated for 2-1/2 hours at nominal voltage and shall be de-energized for 1/2 hour. This duty cycle shall be conducted at least eight times each day. Time counted toward the reliability test shall be the sum of all 2-1/2 hour running periods.

Following the initial hot and cold tests of the one temperature cycle (see 4.4.3.2.1), the tests shall continue at the test level of para. 4.4.3.2. At all times when power is applied during the Reliability test, including heat and cold (except for a period including the starting cycle on each start and not to exceed 5 minutes) the displacement gyro, the rate switching gyro, and a Scorsby table operating at between 5 and 7 cpm and at an angle sufficient to cause positive switching of the rate switching gyro.

*4.4.3.5 Performance Characteristics to be Measured - The performance characteristics to be measured shall be as specified in 6.10.

*4.4.3.6 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the Acceptance requirement (paragraph 4.4.3.5) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

*4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

*4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

*4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

*4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

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(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

*4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room
Altitude	Normal ground (0 - 5000 ft.)
Humidity	Room ambient
A.C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D.C. Voltage	27.0 ± 2.0 volts

*4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

*4.5.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

*4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number
(2) The circuit reference symbol number
(3) The part function
(4) Name of the manufacturer
(5) Nature of the failure
(6) The number of hours which the part operated prior to failure

*4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In

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this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

*4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

*4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-R-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

*4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

*4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the government inspector.

5. PREPARATION FOR DELIVERY

*5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Intended Use - The attitude heading reference system defined by this specification is intended for use as an all-attitude heading and reference system for aircraft. The system provides roll, pitch, and heading signals to the pilot's indicators and to other avionic equipment.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptance standard of performance.

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*6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

*6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

*6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Reference Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

*6.6 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

*6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

(3) Data requirements (see 3.6).

6.8 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

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6.9 Items of Equipment - The equipment covered by this specification shall consist of the following items:

Item	Type Designation	Appl. Par.
Gyroscope Assembly, Displ.	CN-1054/A	3.4.1
Amplifier-Power Supply	AM-4274/ASN-50	3.4.2
Adapter-Compensator, Compass	MX-6985/ASN-50	3.4.3
Controller, Compass System	C-2984/AJB-3A	3.4.4
Gyroscope, Rate Switching	CN-495A/AJB-3	3.4.5

6.9.1 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment:

Item	Type Designation
Transmitter, Remote Compass	ML-1

*6.10 Performance Characteristics to be Measured -

Function	Accept Performance
Roll, Pitch, Heading Accuracy (At each 45° heading)	Repeat platform within ± 1°
Dynamic Operation (About all three axes)	Hunting or jumping not to exceed 1°
Direction Gyro Drift	Not to exceed 3° per hour when corrected for earth's rate
Heading Drift	Not to exceed 3° per hour with azimuth compensation but no correction for earth rate
Vertical Gyro Drift (with spin axis deflected 20°)	Not to exceed 25° per hour when corrected for earth's rate
Mode Operation	Synchronize within ± 3°

*6.11 Applicable Documents -

SPECIFICATIONS

Military

MIL-E-4682	Tubes and Transistors, Choice and Application of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for

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MIL-T-5422	Testing, Environmental Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-P-7788	Plate, Plastic, Lighting
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-M-18012	Markings for Aircrew Station Displays, Design and Configuration of
MIL-T-18303	Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-19576	Transmitter, Remote Compass, Thin Wing, Type ML-1 (unstabilized)
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for
MIL-G-23723	Gyroscope, Rate Switching CN-495A/AJB-3

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MS17322	Meter, Time, Totalizing, Miniature 400 Cycle, Digital

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MS17323	Meter, Time, Totalizing, Miniature 400 Cycle, Dial
MS17392	Controller, Compass System C-2984/AJB-3A

*6.11.1 Availability of Documents -

(1) When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

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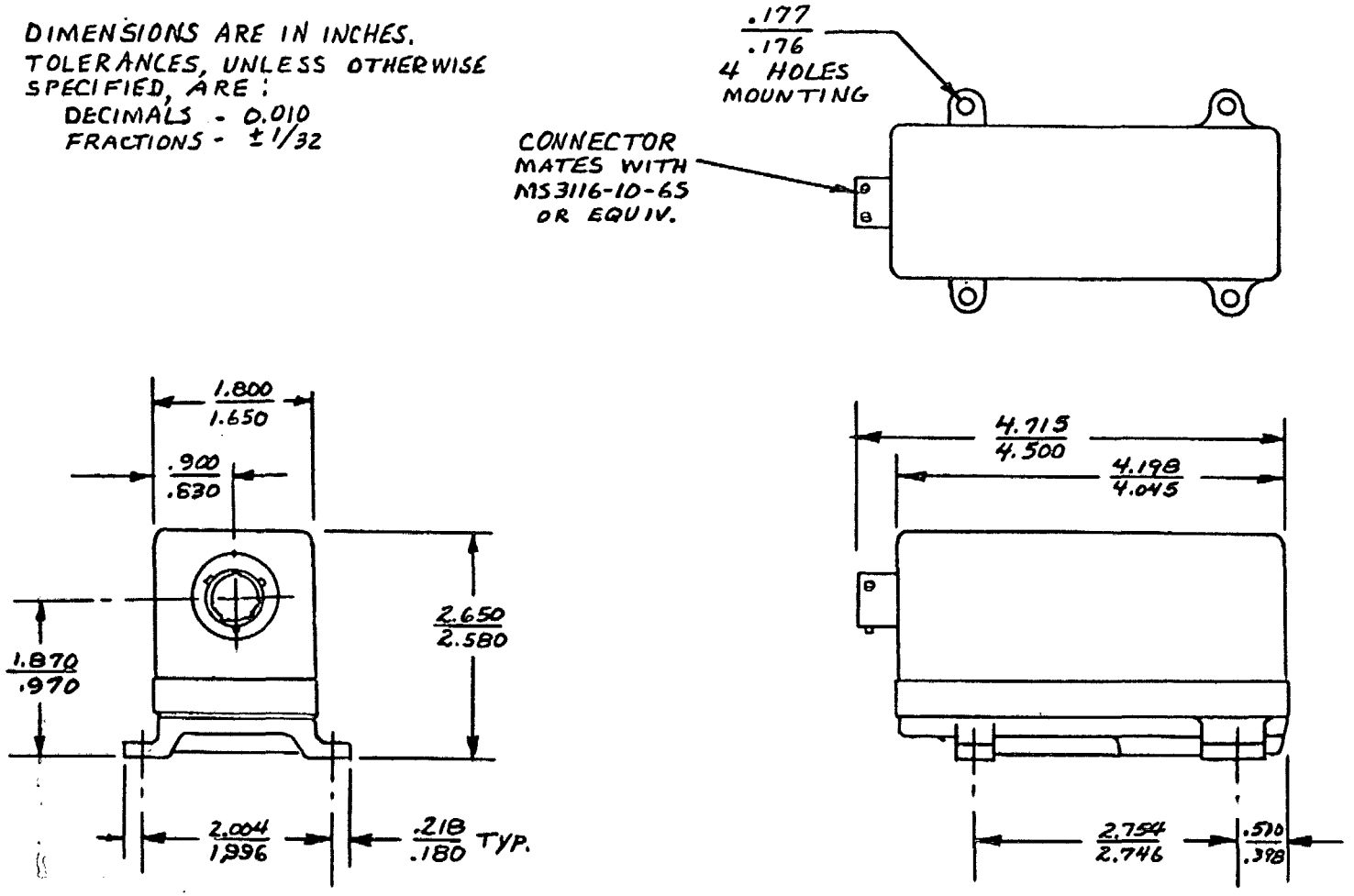


FIGURE 6 RATE SWITCHING GYROSCOPE

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-A-23670A(WP) ATTITUDE HEADING REFERENCE SYSTEM AN/ASN-50(*)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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